Design data security from classification to the AI index



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Checking in with Globomantics



Two regulators, one set of formulas

One wants your keys, kept in-region

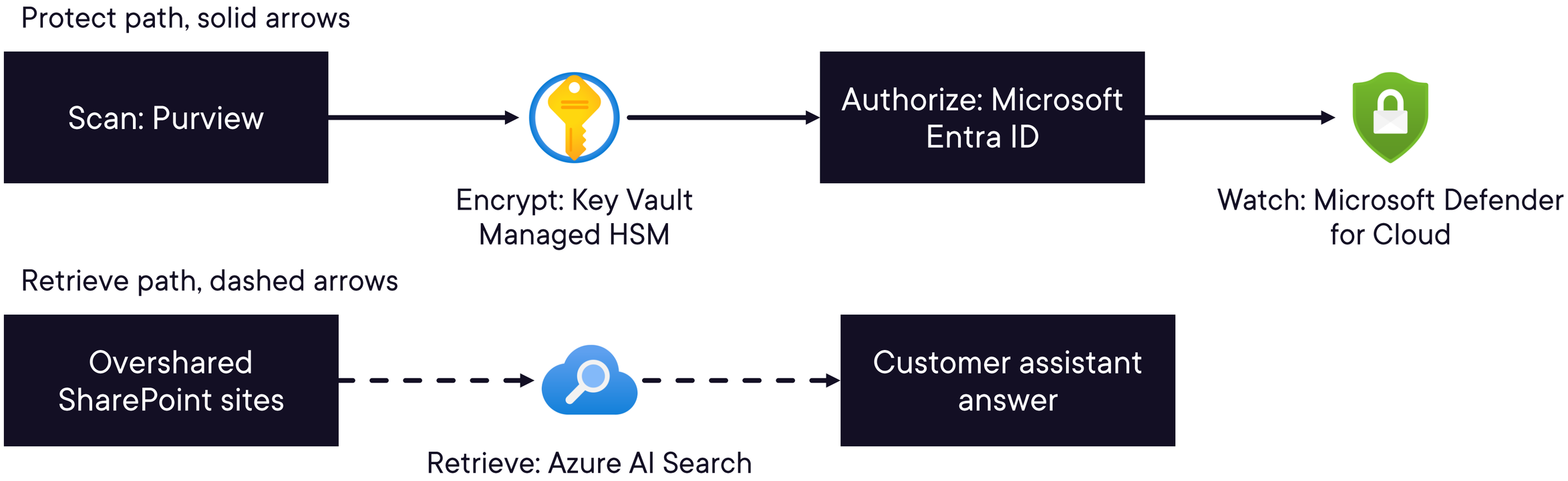
The other wants records searchable

Encrypt first, or find the data first?

Pick the discovery tool by scope

Tool	Finds	Gives you
Purview Data Map	Any cloud, plus on-premises	Metadata and lineage only
Purview Information Protection	Sensitive content in files and mail	Labels that travel with the content
SQL data discovery and classification	Columns in a single database	Column labels that feed auditing
Defender CSPM sensitive data discovery	Which stores hold sensitive data	Risk context, nothing labeled

Follow a formulation record to the answer



Customer-controlled keys protect the record, and the index repeats every overshare

Specify priorities for mitigating threats to data

Order the protections like the benchmark

Classify first, with DP-1

Threats reorder the middle

Let Defender rank the risk

Customer-managed keys last

Evaluate encryption with Azure Key Vault and Managed HSM

Let the regulator pick key custody

Option	Root of trust	Choose it when
Platform-managed keys	Microsoft	No rule names custody
Key Vault Premium	Microsoft, multitenant HSM	You need your own key
Key Vault Managed HSM	Your security domain	A rule says single-tenant
Azure Cloud HSM	You, as infrastructure	IaaS apps, not PaaS encryption

Add layers without moving custody

Infrastructure encryption

TLS 1.2 or later for data in transit

Encryption at host for new VMs

Secure data in Azure SQL, Azure Synapse, and Azure Cosmos DB

Pick the SQL control by who mustn't read it

Transparent data encryption	vs.	Always Encrypted
Protects files, logs, and backups		Keys stay with the client app
The server still reads plaintext		Administrators can't read values
On by default for new databases		Secure enclaves for richer queries

Close the data plane in Synapse and Cosmos DB



Azure Synapse Analytics

- Managed virtual network
- Data exfiltration protection

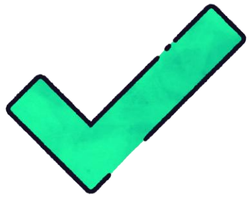


Azure Cosmos DB

- Data-plane role-based access
- disableLocalAuth set to true

Secure data in Azure Storage

Disallow Shared Key on every account





Do

Sign with a Microsoft Entra ID token: Conditional Access applies

Use a user delegation SAS: Entra ID signs it

Don’t

Rely on the account key: rejected with a 403

Hand out an account or service SAS: the key signs it

Finish the storage account design

Watch who can list the keys

Lock retention with immutable storage

Private endpoints, no public blob access

Select Defender for Storage and Defender for Databases

Watch the identities you allowed

Defender for Storage	vs.	Defender for Databases
Activity monitoring		Azure SQL and SQL Servers
Malware scanning, billed per GB		Open-source relational databases
Sensitive data threat detection		Azure Cosmos DB, NoSQL API only

Design security for data used in AI workloads

Indexes inherit permissions

Trimming copies the source permissions

Fix oversharing before you index

Label enforcement is still in preview



Know whose token reaches the data

A user's own token	vs.	An agent's own identity
Search trims for that user		No per-user trimming
Purview policies can act		Scoped by role assignments
Labels checked per caller		Audited, not enforced

Demo: Audit the customer assistant's data path

Its storage account, read with the Azure CLI, then the oversharing assessment for the sites its index crawled

Practice question and key points

Theme: Key custody for data at rest

Question

Globomantics has an Azure subscription. The Materials business unit stores the data of each customer in a separate Azure Storage account. Each customer contract requires Globomantics to make the data of the customer unreadable when the contract ends.

You need to recommend an encryption solution that meets the following requirements:

- **Make the data of a single customer unreadable on demand.**
- **Minimize costs.**

Theme: Key custody for data at rest

Question

You need to recommend an encryption solution for the storage accounts. What should you include in the recommendation?

A. customer-managed keys in Azure Key Vault Standard

B. customer-managed keys in Azure Key Vault Managed HSM

C. infrastructure encryption for the storage accounts

D. platform-managed keys for the storage accounts

Theme: Key custody for data at rest

Answer: A

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Exam tips for SC-100

Name the actor first

Find the deciding word

Know GA from preview

Practice before exam day

Checking out with Globomantics



Find and classify the data first

Let the regulator pick key custody

Move every store to Entra ID

Fix oversharing before you index

Thank you for your time!



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